

Obtenha a matriz inversa, se existir, de:

1. $A = \begin{pmatrix} 1 & -1 \\ 3 & 1 \end{pmatrix}$

2. $A = \begin{pmatrix} 5 & 7 \\ 2 & 3 \end{pmatrix}$

3. $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$

4. Dada $A = \begin{pmatrix} 2 & m \\ -m & 1/2 \end{pmatrix}$, calcule m de modo que se tenha $A^{-1} = A^t$.

5. Dada $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, qual é a matriz A^2 ? A^{411} ?

(1)

$$A = \begin{bmatrix} 1 & -1 \\ 3 & 1 \end{bmatrix} \quad A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad A^{-1} = \frac{1}{1 \times 1 - (-1) \times 3} \begin{bmatrix} 1 & 1 \\ -3 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{1 - (-3)} \begin{bmatrix} 1 & 1 \\ -3 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 1 & 1 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} \\ -\frac{3}{4} & \frac{1}{4} \end{bmatrix} \quad A^{-1}$$

$$A \times A^{-1} = I_2$$

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & -1 \\ 3 & 1 \end{bmatrix} \times \begin{bmatrix} \frac{1}{4} & \frac{1}{4} \\ -\frac{3}{4} & \frac{1}{4} \end{bmatrix} = \begin{bmatrix} 1 \times \frac{1}{4} + (-1) \times (-\frac{3}{4}) & 1 \times \frac{1}{4} + (-1) \times \frac{1}{4} \\ 3 \times \frac{1}{4} + 1 \times (-\frac{3}{4}) & 3 \times \frac{1}{4} + 1 \times \frac{1}{4} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

(2)

$$A = \begin{bmatrix} 5 & 7 \\ 2 & 3 \end{bmatrix} \quad A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad A^{-1} = \frac{1}{5 \times 3 - 7 \times 2} \begin{bmatrix} 3 & -7 \\ -2 & 5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{15-14} \times \begin{bmatrix} 3 & -7 \\ -2 & 5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{1} \times \begin{bmatrix} 3 & -7 \\ -2 & 5 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 3 & -7 \\ -2 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 5 \times 3 + 7 \times (-2) & 5 \times (-7) + 7 \times 5 \\ 2 \times 3 + (-2) \times (-2) & 2 \times (-7) + 3 \times 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A \times A^{-1} = I_2$$

$$\begin{bmatrix} 5 & 7 \\ 2 & 3 \end{bmatrix} \times \begin{bmatrix} 3 & -7 \\ -2 & 5 \end{bmatrix} =$$

(3)

$$A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$(1 \times 1) - (-1 \times -1) = 0$$

Não é invertível

(5)

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 \times 1 + 0 \times 0 & 1 \times 0 + 0 \times (-1) \\ 0 \times 1 + (-1) \times 0 & 0 \times 0 + (-1) \times (-1) \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A^1 = A$$

$$A^2 = I$$

$$A^3 = A$$

$$A^4 = I$$

$$A^{77} = A$$

$$A^{77} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$